

$$\sum_{i=1}^n \frac{1}{i^2} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2} = ?$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots$$

$$\frac{\sin x}{x} = 1 - \frac{x^2}{6} + \frac{x^4}{120} - \dots$$

$$\frac{\sin x}{x} = 0 \quad \text{корни } \sin x \text{ в } x = \pm\pi, \pm2\pi, \pm3\pi, \dots$$

$$\frac{\sin x}{x} = \left(1 - \frac{x}{\pi}\right) \left(1 + \frac{x}{\pi}\right) \left(1 - \frac{x}{2\pi}\right) \left(1 + \frac{x}{2\pi}\right) \dots = \prod_{n=1}^{\infty} \left(1 - \frac{x^2}{n^2\pi^2}\right)$$

$$\text{при } x^2 \text{ коэф} = -\frac{1}{6}$$

$$\text{если берем } n \neq \infty \Rightarrow 1 - \frac{x^2}{\pi^2} - \frac{x^2}{4\pi^2} - \dots = \left| -\frac{x^2}{6} \right| \cdot \frac{\pi^2}{x^2}$$

краве 1-ю

$$-\frac{1}{\pi^2} \left(1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots\right) = -\frac{1}{6}$$

$$\left(1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots\right) = \frac{\pi^2}{6} \quad \mathbb{Z}$$

• In the Product: To get an  $x^2$  term when multiplying all those infinite brackets, he had to take the "1" from all but one bracket, and the  $-\frac{x^2}{n^2\pi^2}$  term from exactly one bracket. So, the  $x^2$  coefficient is the sum of all those individual terms:

Задача 1 #125938 Максимум баллов за задание: 4

Найдите все значения параметра  $a$ , при каждом из которых уравнение

$$(4x - 3|x + a^2| + |x - 1| + 3a^2)^2 - (a + 1)(4x - 3|x + a^2| + |x - 1| + 3a^2) + 4 = 0$$

имеет ровно два различных решения.

$$1) y = 4x - 3|x + a^2| + |x - 1| + 3a^2$$

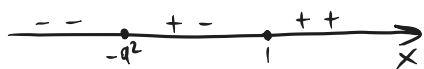
$$y^2 - (a + 1)y + 4 = 0$$

$$x + a^2 = 0 \quad x - 1 = 0$$

$$x = -a^2 \leq 0 \quad \underline{x = 1}$$

но определено  
(квадратное уравнение)  $\leq 0$

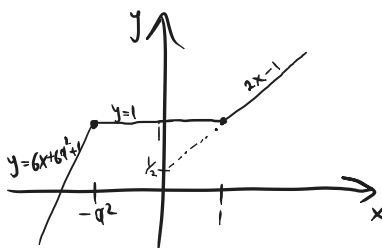
модули:



$$2) \text{ если } x \leq -a^2 \Rightarrow y = 4x - 3x + 3a^2 - (x - 1) + 3a^2 = 6x + 6a^2 + 1$$

$$\text{если } -a^2 < x < 1 \Rightarrow y = 4x - 3x - 3a^2 - x + 1 + 3a^2 = 1$$

$$\text{если } x \geq 1 \Rightarrow y = 4x - 3x - 3a^2 + x - 1 + 3a^2 = 2x - 1$$



$$3) \text{ если } y = 1 \Rightarrow y = 4x - 3|x + a^2| + |x - 1| + 3a^2 \text{ имеет 20 решений}$$

$$y \neq 1 \Rightarrow \text{ровно 1 решение}$$

связь по графику

$$4) f(y) = y^2 - (a + 1)y + 4$$

$$\begin{cases} D > 0 \\ f(0) > 0 \end{cases} \Rightarrow \begin{cases} (a+1)^2 - 4 > 0 \\ 1 - (a+1) + 4 > 0 \end{cases} \Rightarrow \begin{cases} (a-3)(a+5) > 0 \\ a \neq 4 \end{cases} \Rightarrow \begin{cases} a \in (-\infty, -5) \cup (3, +\infty) \\ a \neq 4 \end{cases}$$

Найдите все значения параметра  $a$ , при каждом из которых уравнение

$$(|x - a^2| + |x + 1|)^2 - 7(|x - a^2| + |x + 1|) + 4a^2 + 4 = 0$$

имеет ровно два различных корня.

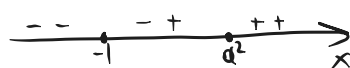
$$y = |x - a^2| + |x + 1|$$

$$f(y) = y^2 - 7y + 4a^2 + 4 = 0$$

1) нули модулей

$$x - a^2 = 0 \quad x + 1 = 0$$

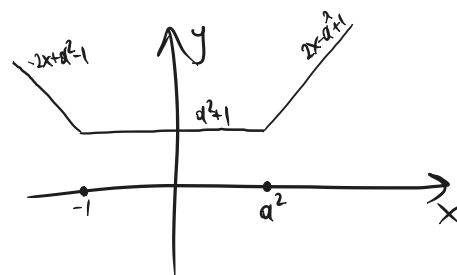
$$\underline{x = a^2 \geq 0} \quad \underline{x = -1}$$



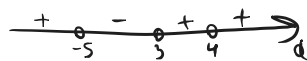
$$2) \text{ если } x \leq -1 \Rightarrow y = -x + a^2 - x - 1 = -2x + a^2 - 1$$

$$\text{если } -1 < x < a^2 \Rightarrow y = -x + a^2 + x + 1 = a^2 + 1$$

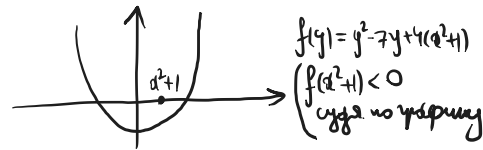
$$\text{если } x \geq a^2 \Rightarrow y = x - a^2 + x + 1 = 2x - a^2 + 1$$



Antwort:  $a \in (-\infty, -5) \cup (3, 4) \cup (4, +\infty)$



3) wenn  $y = d^2 + 1 \Rightarrow \infty$  prim- und  
wenn  $y > d^2 + 1 \Rightarrow 2$  prim- und  
wenn  $y < d^2 + 1 \Rightarrow \emptyset$



$$\frac{49}{16} > d^2 + 1$$

$$d^2 < \frac{49}{16} - 1 = \frac{33}{16}$$

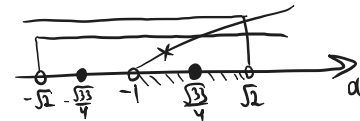
$$\sqrt{\frac{33}{16}} < \sqrt{2}$$

Чтобы уравнение  $y^2 - 7y + 4a^2 + 4 = 0$  имело один корень, необходимо потребовать  $D = 0$ :

$$4) f(y) = y^2 - 7y + 4a^2 + 4 = 0$$

$$\begin{cases} D=0 \\ f(q^{2+1}) > \cancel{0} \\ < 0 \end{cases} \Rightarrow \begin{cases} z^2 - q^2(q^{2+1}) = 0 \\ (q^{2+1})^2 - 7(q^{2+1}) + 4(q^{2+1}) > \cancel{0} \\ < 0 \end{cases}$$

$$\left\{ \begin{array}{l} |a| = \sqrt{\frac{33}{16}} \\ (a^2+1)(a^2-3) > 0 \\ \begin{array}{ccc} > 0 & < 0 & < 0 \\ < 0 & > 0 & \end{array} \end{array} \right.$$



$$Q \in (\pm \frac{\sqrt{33}}{4}) \cup (-\sqrt{2}, \sqrt{2})$$

$\Rightarrow \begin{cases} a^2 > -1 \\ a^2 < 2 \end{cases} \text{ where } \begin{cases} a^2 < -1 \\ a^2 > 2 \end{cases}$   
 $\Rightarrow \begin{cases} 1 < a^2 < 2 \\ a^2 < -1 \text{ or } a^2 > 2 \end{cases} \Rightarrow \emptyset$

Найдите все значения параметра  $a$ , при каждом из которых уравнение

$$(4x + |x - a| - |3x + 1|)^2 - (a + 1)(4x + |x - a| - |3x + 1|) + 1 = 0$$

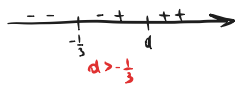
имеет ровно два различных решения.

$$y = 4x + |x - 1| - 13x + 11$$

$$f(y) = y^2 - (a+1)y + 1 = 0$$

1) Изм. модифици

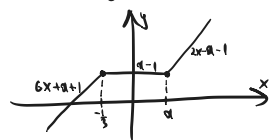
$$1^o \quad \begin{array}{l} x - d = 0 \\ x = d \end{array} \quad \begin{array}{l} 3x + 1 = 0 \\ x = -\frac{1}{3} \end{array}$$



2) wenn  $x \leq -\frac{1}{3} \Rightarrow y = 4x - x + a + 3x + 1 = 6x + a + 1$

case  $-\frac{1}{3} < x < a \Rightarrow y = 4x - x + a - 3x - 1 = a - 1$

also  $x \geq a \Rightarrow y = 4x + x - a - 3x - 1 = 2x - a - 1$


$$2\rho_{\text{row-vec}, \text{case D}} > 0$$

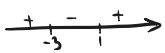
$$3) y \neq a-1 \Rightarrow \exists 1 \text{ plem-ue}$$

$$D > 0$$

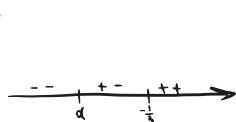
$$\begin{cases} D > 0 \\ f(a-1) \neq 0 \end{cases} \quad \begin{cases} (a+1)^2 - 4 > 0 \\ (a-1)^2 - (a-1)(a+1) + 1 \neq 0 \end{cases}$$

$$\begin{cases} (a+1)^2 - 1^2 > 0 \\ (a-1)(a-1-(a+1)) \neq -1 \end{cases} \quad \begin{cases} (a+1+2)(a+1-2) > 0 \\ a \neq \frac{3}{2} \end{cases}$$

$$\begin{aligned} -2(a-1) &\neq -1 \quad | \cdot (-1) \\ 2a-2 &\neq 1 \\ a &\neq \frac{3}{2} \end{aligned} \quad \left\{ \begin{aligned} (a+3)(a-1) &> 0 \\ a &\neq \frac{3}{2} \end{aligned} \right.$$



$$a \in (-\infty, -3) \cup (1, \frac{3}{2}) \cup (\frac{3}{2}, +\infty) \leftarrow \text{Ansbem}$$

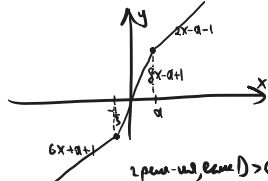


$$d < -\frac{1}{3}$$

2) wenn  $x \leq d \Rightarrow y = 4x - x + d + 3x + 1 = 6x + d + 1$

case  $-\frac{1}{3} < x < a \Rightarrow y = 4x + x - a + 3x + 1 = 8x - a + 1$

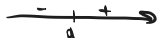
even  $x \geq a \Rightarrow y = 4x + x - a - 3x - 1 = 2x - a - 1$


$$2\rho_{\text{new-old}} \cos \theta > 0$$

3)  $D > 0$

$$(a+1)^2 - 4 > 0$$

$$a \in (-\infty, -3) \cup (1, +\infty)$$

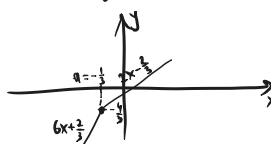


$$y = 4x + |x-1| - |3x+1| = 4x + |x + \frac{1}{3}| - \underbrace{|3x+1|}_{3|x+\frac{1}{3}|} = 4x - 2|x + \frac{1}{3}|$$

also  $x > a \Rightarrow y = 2x - \frac{2}{3}$

also  $x < a \Rightarrow y = 6x + \frac{2}{3}$

Case  $x = a = -\frac{1}{3} \Rightarrow y = 4x = -\frac{4}{3}$



решено 1 реш.-ие